

Experiment No: 5

Title: Write a Python Program for the Missionaries and Cannibals Problem

Aim

To develop a Python program to solve the **Missionaries and Cannibals Problem** by safely transporting all missionaries and cannibals across the river without violating the problem constraints.

Objective

To implement the Missionaries and Cannibals problem in Python by generating valid boat movements and ensuring that the number of cannibals never exceeds the number of missionaries on either riverbank.

Algorithm

- **Step 1:** Start the program.
- **Step 2:** Define the initial state and the goal state.
- **Step 3:** Generate all possible valid boat moves.
- **Step 4:** Check whether each new state is valid.
- **Step 5:** If the state is valid and not visited, add it to the queue.
- **Step 6:** Repeat the process until the goal state is reached.
- **Step 7:** Display the sequence of valid states.
- **Step 8:** Stop the program.

Program

```
from collections import deque
```

```
# Check whether the state is valid
```

```
def is_valid(m_left, c_left):
```

```
    m_right = 3 - m_left
```

```
    c_right = 3 - c_left
```

```
    if m_left < 0 or c_left < 0 or m_left > 3 or c_left > 3:
```

```
        return False
```

```
    if (m_left > 0 and c_left > m_left):
```

```
return False
```

```
if (m_right > 0 and c_right > m_right):
```

```
    return False
```

```
return True
```

```
# BFS Algorithm
```

```
def missionaries_cannibals():
```

```
    start = (3, 3, 'L')
```

```
    goal = (0, 0, 'R')
```

```
    moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]
```

```
    queue = deque([[start]])
```

```
    visited = set()
```

```
    while queue:
```

```
        path = queue.popleft()
```

```
        state = path[-1]
```

```
        if state == goal:
```

```
            print("Solution Path:\n")
```

```
            for s in path:
```

```
                print(s)
```

```
            return
```

```
        if state in visited:
```

```
            continue
```

```

visited.add(state)

m, c, boat = state

for dm, dc in moves:

    if boat == 'L':
        new_state = (m-dm, c-dc, 'R')
    else:
        new_state = (m+dm, c+dc, 'L')

    if is_valid(new_state[0], new_state[1]):
        queue.append(path + [new_state])

print("No Solution Found")

missionaries_cannibals()

```

Sample Input

No user input required.

Sample Output

Solution Path:

```

(3, 3, 'L')
(3, 1, 'R')
(3, 2, 'L')
(3, 0, 'R')
(3, 1, 'L')
(1, 1, 'R')
(2, 2, 'L')
(0, 2, 'R')
(0, 3, 'L')
(0, 1, 'R')
(1, 1, 'L')
(0, 0, 'R')

```

Result

Thus, the Python program for the **Missionaries and Cannibals Problem** was executed successfully, and all missionaries and cannibals were safely transported across the river without violating the given constraints.